

Architectural Evolution of nopCommerce

Scenario C - Omnichannel Commerce Core

TP1 Group 02

Presented by:

- António Alberto 114622
 - Diogo Fernandes 114137
 - João Roldão 113920
 - João Varela 113780
-

SCENARIO CHOICE

The Problem

VerdeMart uses nopCommerce as its web storefront only

Order state needs to reflect stores and warehouses

Operations cannot stop when external systems lag

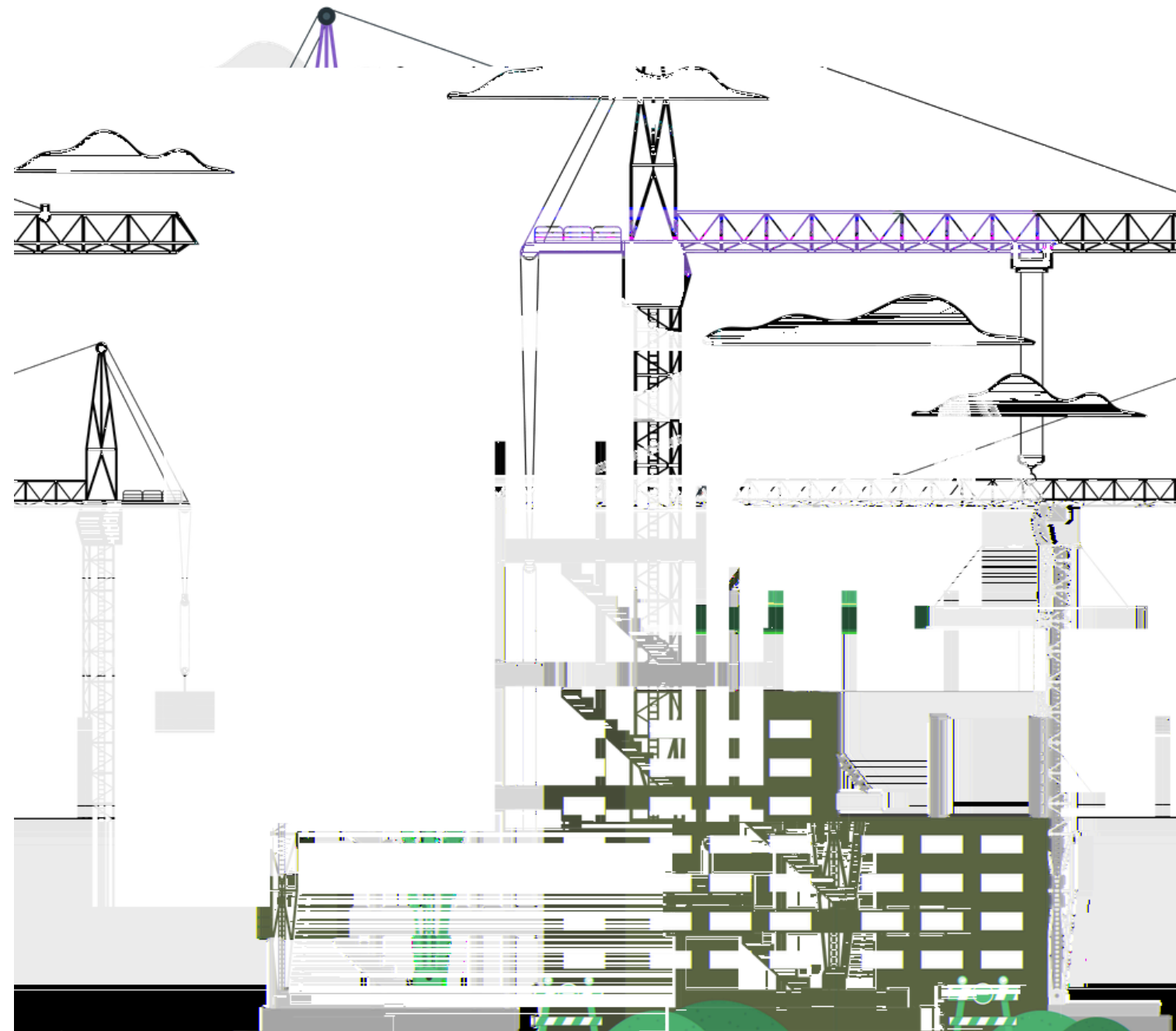
The Choice

Map directly onto existing nopCommerce domain

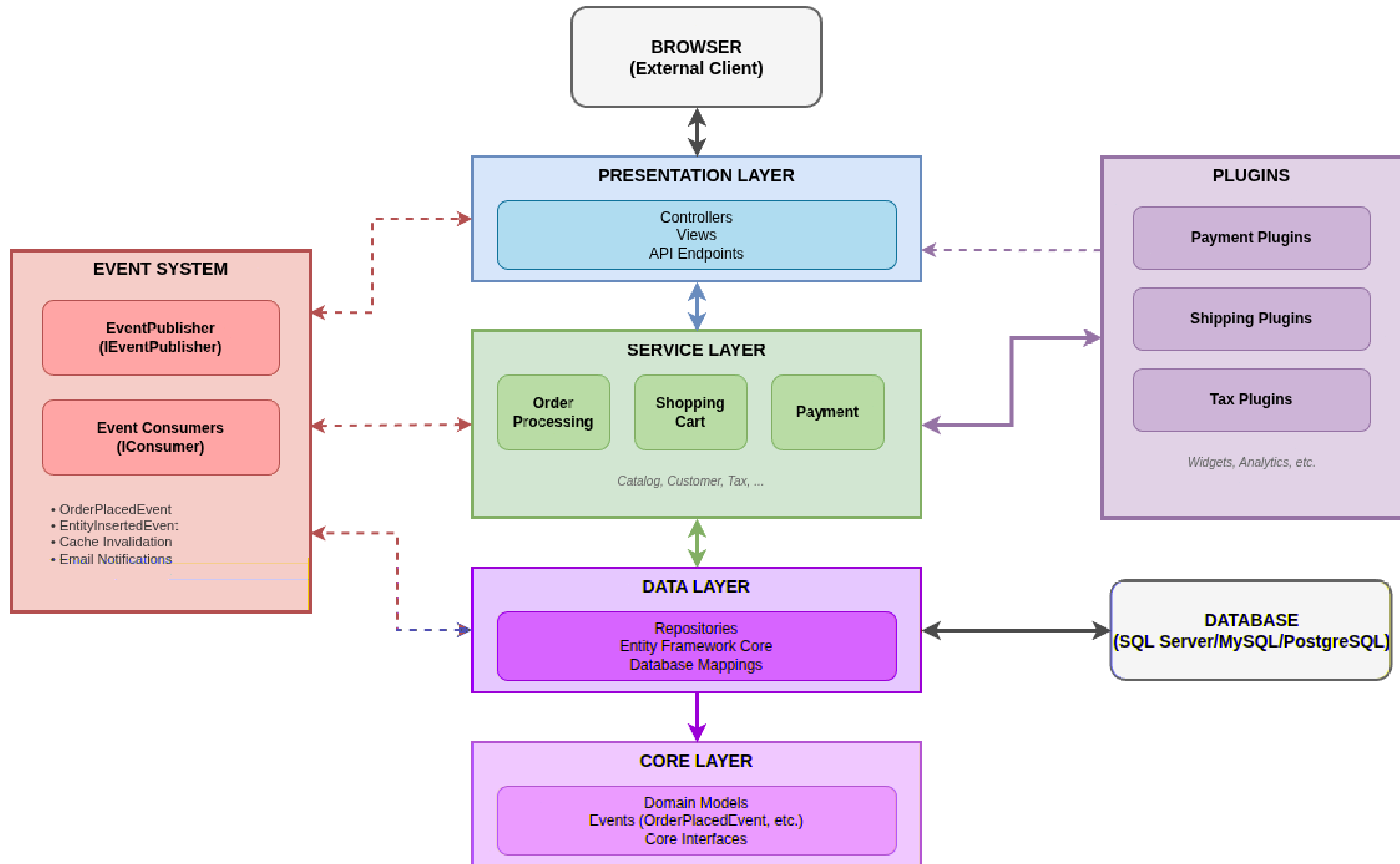
Turn nopCommerce into a Commerce Core

Evolve, don't Invent

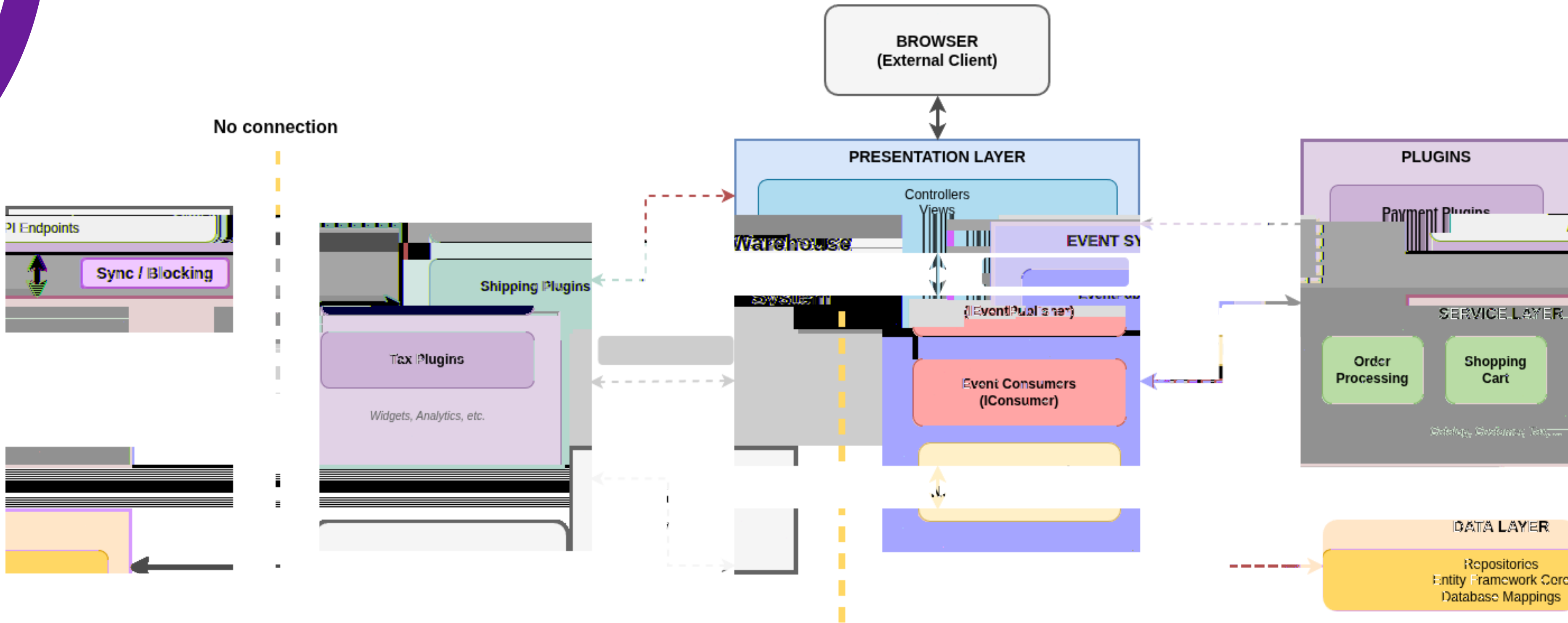
CURRENT STATE ANALYSIS



nopCommerce Communication Architecture



noCommerce Communication Architecture



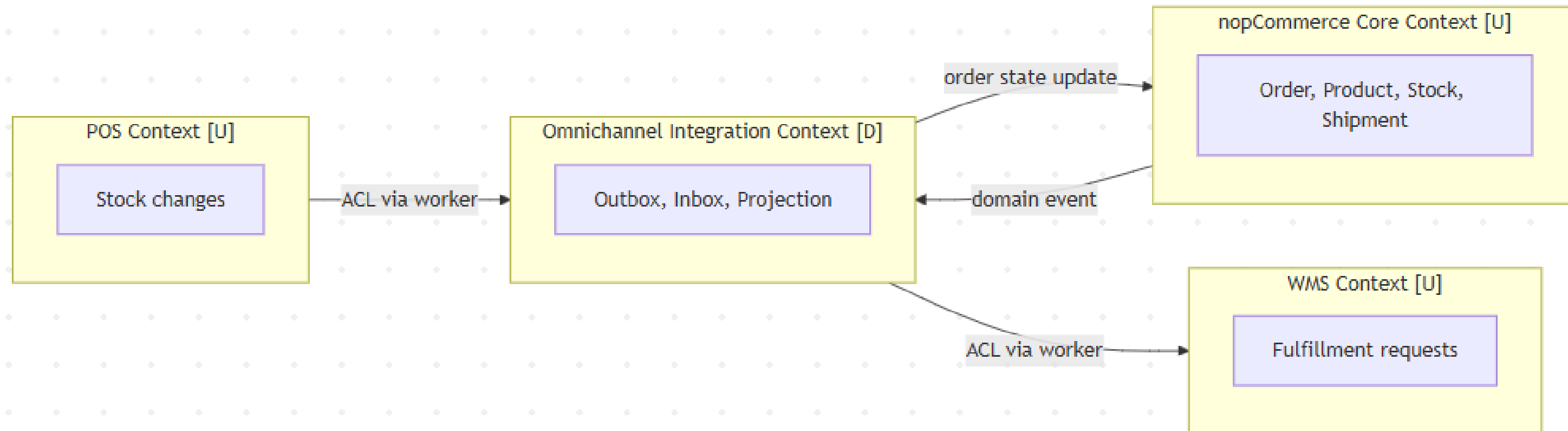
DOMAIN AND BOUNDARY MODEL

6 Subdomains

- **Commerce Core:** checkout, orders, customers, catalog, pricing, payments.
- **Inventory Visibility:** product stock projection, warehouse-specific quantities.
- **Fulfillment Coordination:** bridge from placed order to warehouse processing.
- **Integration Reliability:** outbound, inbound, retries, DLQ, idempotency.
- **Warehouse Operations:** external WMS behavior (accept/reject/delay/confirm).
- **Store Operations / POS:** stock movements have original outside the web channel.

DOMAIN AND BOUNDARY MODEL

4 Bounded contexts



QUALITY ATTRIBUTE SCENARIOS

- **QA-1 Resilience & Recovery:** WMS 503 for 30s → backlog drains ≤ 60 s after recovery & 0 orders pending > 5 min.
- **QA-2 Consistency:** duplicate POS event → rejected ≤ 50 ms & 0 duplicate fulfillments.
- **QA-3 Traceability:** support investigates pending order → outbox → MQ → worker → projection & resolution ≤ 3 admin clicks.
- **QA-4 Performance:** OrderPlacedEvent fired → outbox row written ≤ 100 ms & 0 sync HTTP calls in checkout.
- **QA-5 Cross-Channel Visibility:** WMS marks order shipped → nopCommerce status updated ≤ 10 s & 0 state divergences.

CHOSEN FRAMEWORK

ADD

Attribute-Driven Design

- Iterative top-down decomposition
- Refines the system based on the most important drivers
- One iteration goal at a time

ACDM

Architecture-Centric Design Method

- Evaluations and experiments
- Multiple concerns at the same time
- Go/no-go Decisions
- More Risk-Driven

ADM

Architecture Development Method

- Enterprise oriented method for planning
- Has a Phase-gated cycle
- Suited for Enterprise-wide transformations

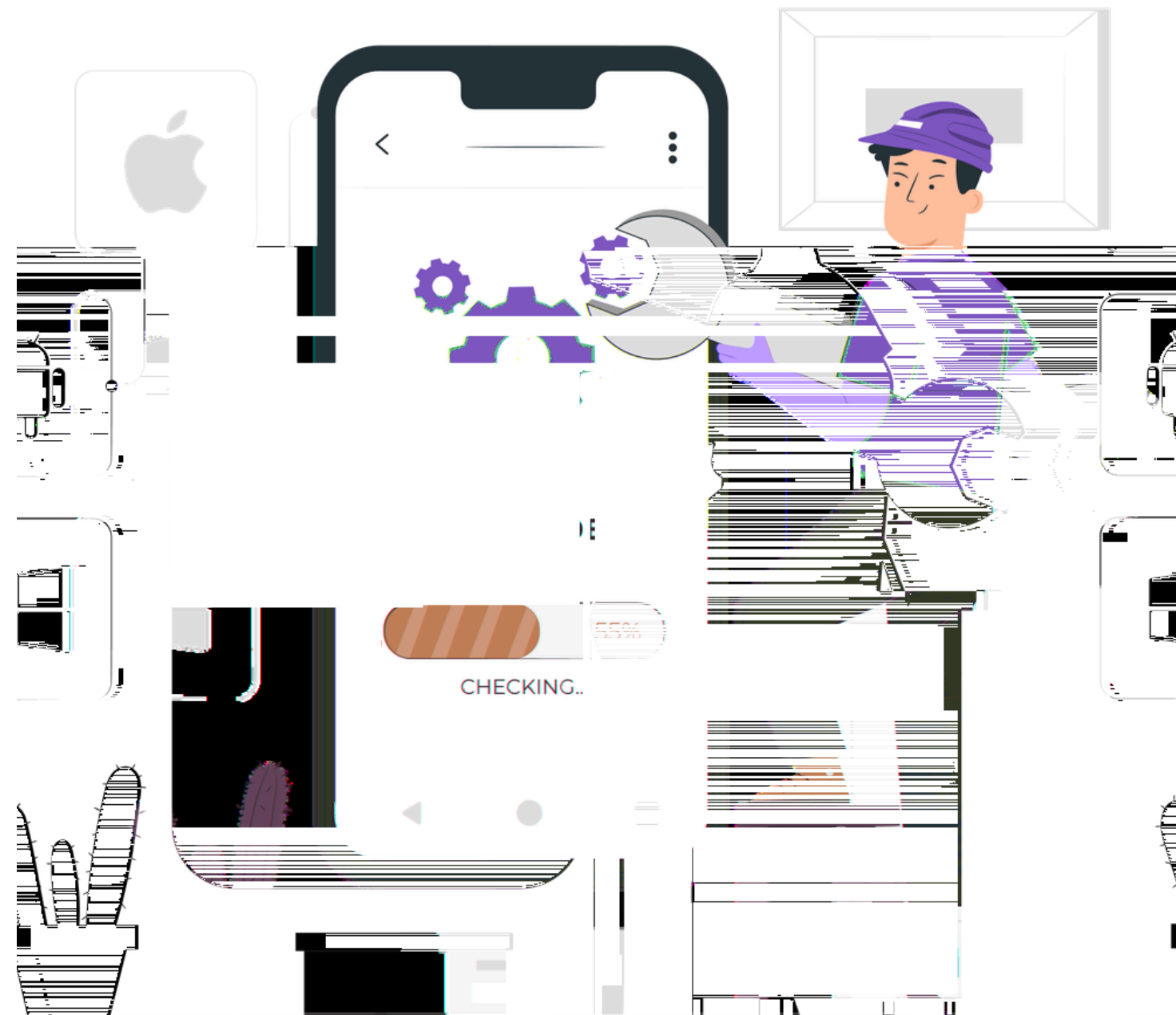
CHOSEN FRAMEWORK

Framework	FIT	Reason

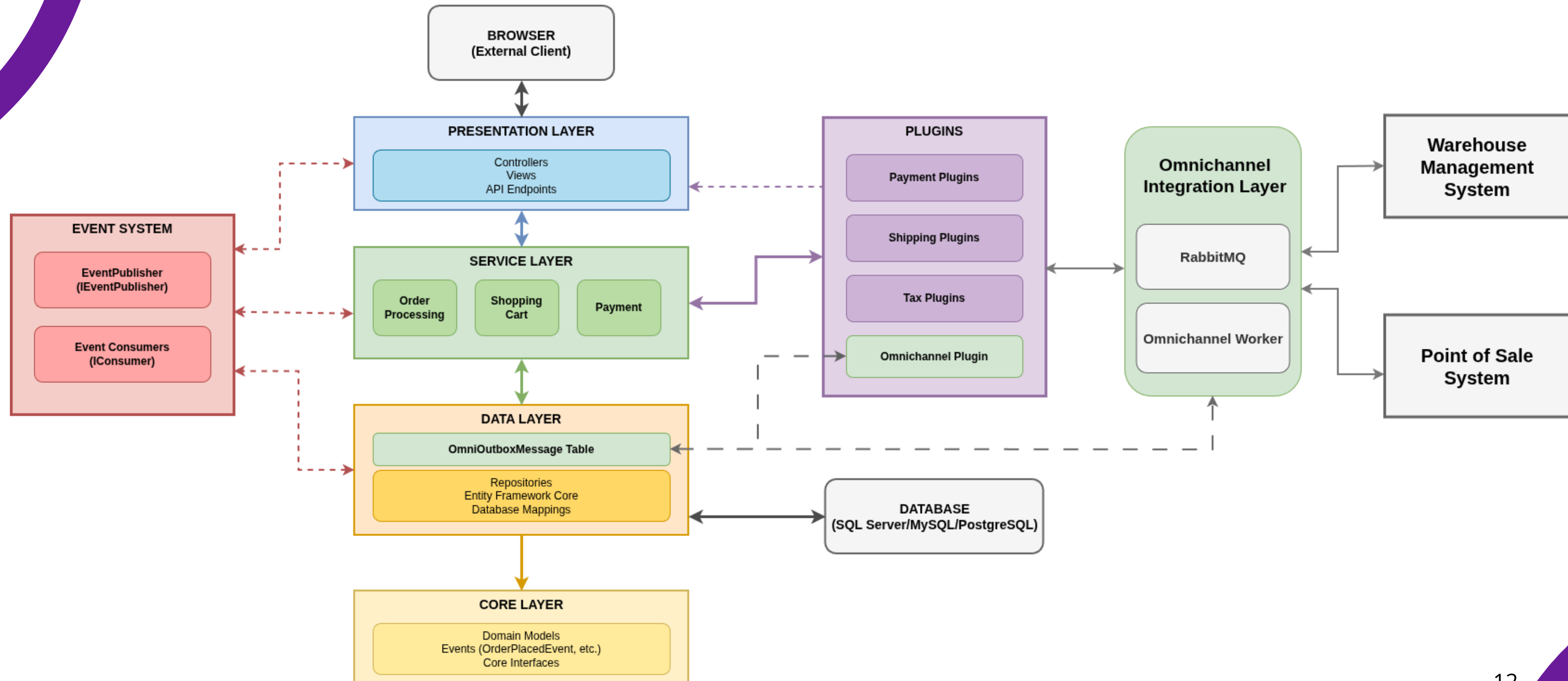
What we use from each

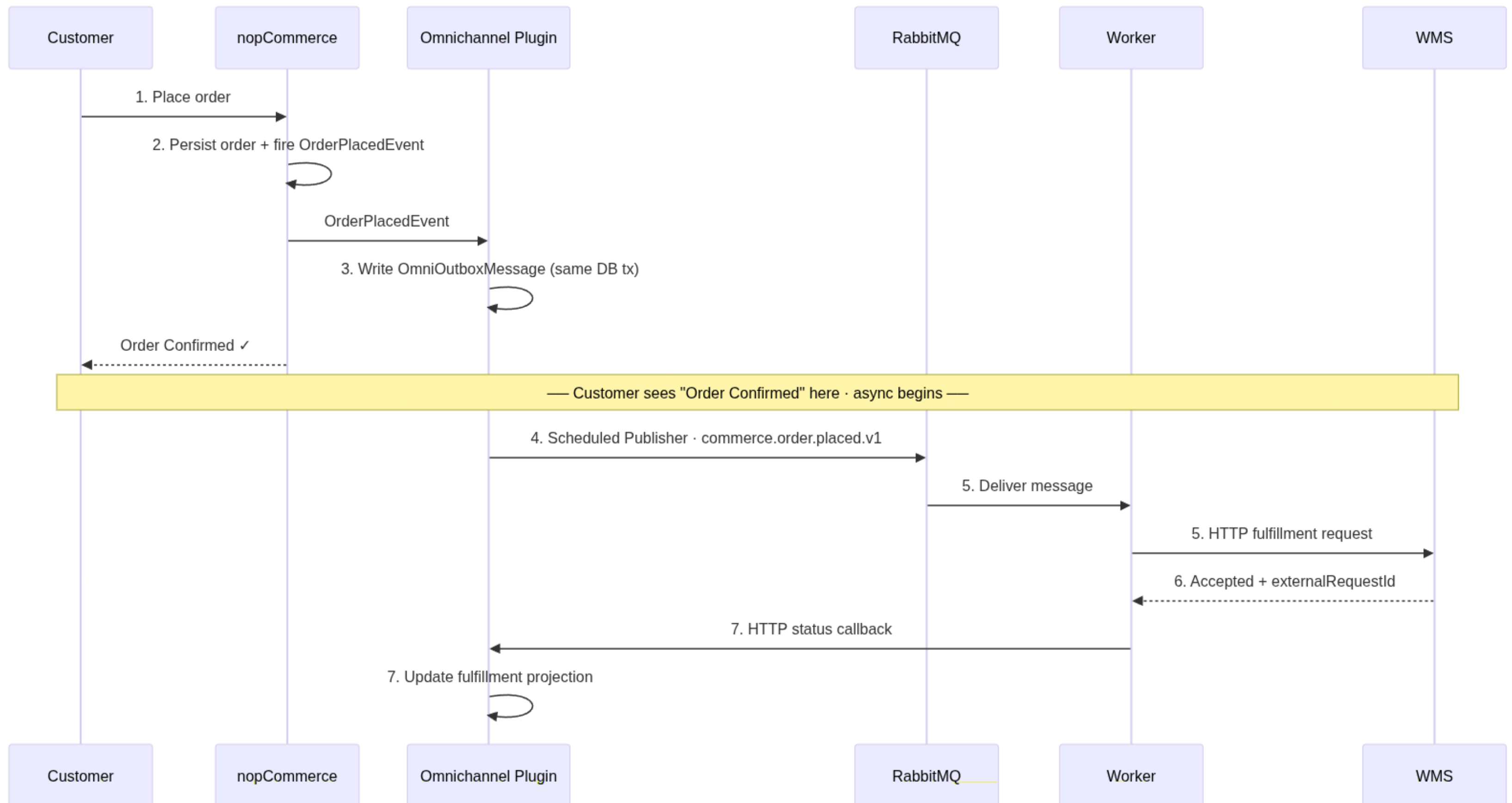
Design Interactions + Tactics
Go / no-go checks
Phase-F Migration Roadmap table

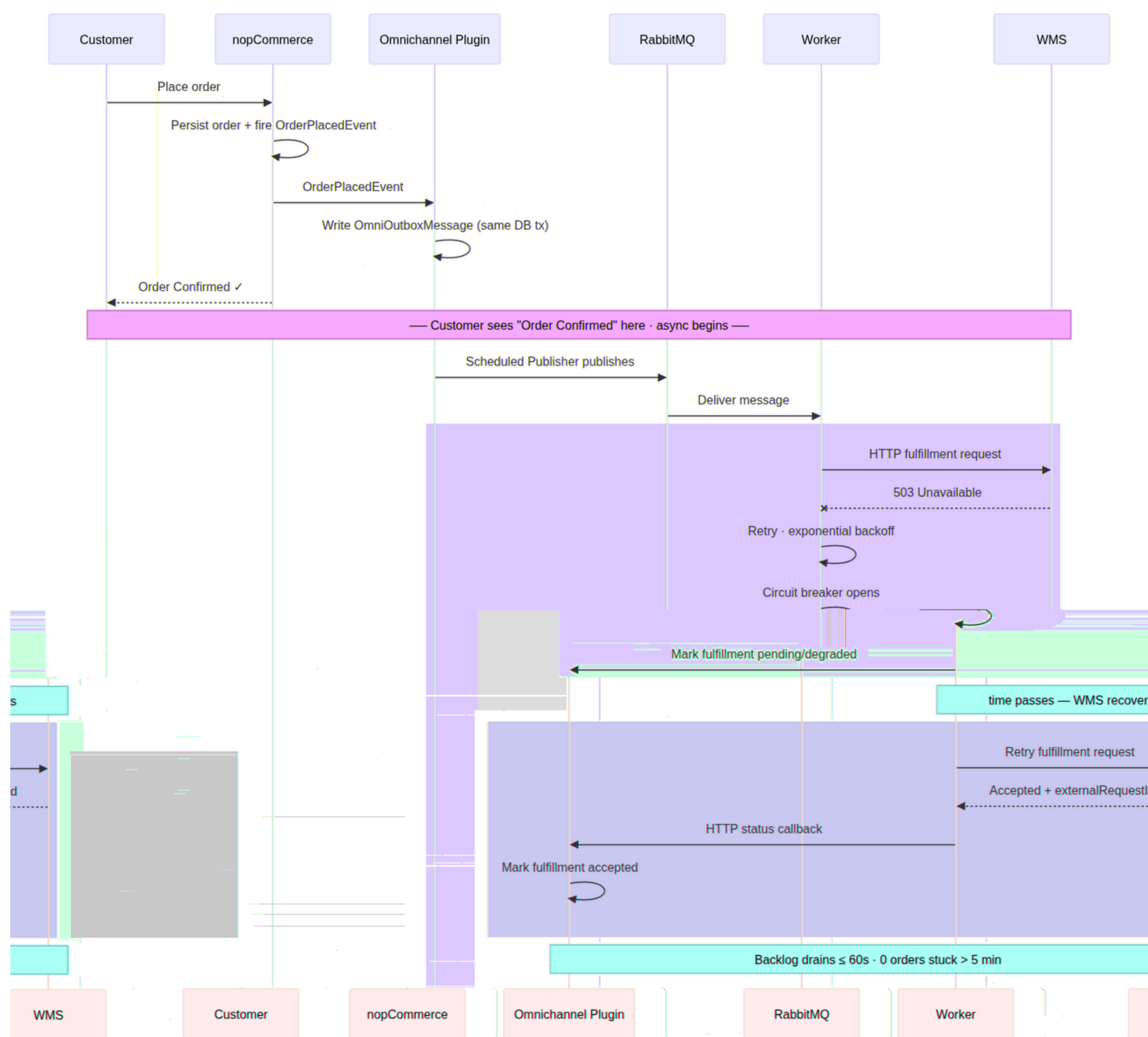
TARGET ARCHITECTURE



nopCommerce Communication Architecture







ARCHITECTURAL DECISIONS

ADR	Decision	Alternative Considered	WHY?
0003	Outbox + RabbitMQ	Sync checkout-to-WMS call	Checkout stays independent from WMS failures
0008	3-ID correlation	Timestamp-only tracing	Delayed orders stay explainable end-to-end
0009	Reject microservices extraction	Extract services by bounded context	Coordination is the architectural problem, decomposition just adds overhead
0010	Reject distributed tracing infra	OpenTelemetry / Jaeger stack	ADR-8's 3-ID correlation satisfies QA-3 with less infrastructure

RISKS & VALIDATION PLAN

Key risks

- Checkout accidentally depends on WMS availability
- Duplicate messages create duplicate fulfillment
- Stock projection becomes misleading
- Scope grows beyond a selective evolution
- Plugin integration is harder than expected

Validation plan

- WMS 503 test + checkout latency
- Replay same message
- Stale POS event + admin drift view
- Roadmap phases + rejected ADRs
- Feasibility spike

FEASIBILITY SPIKE

"Can a placed order start an asynchronous omnichannel workflow without making checkout depend on WMS/POS availability?"

Plugin can write durable outbox row and return.

Checkout stays independent from WMS/POS availability.

EVOLUTION ROADMAP

1. Plugin scaffolding + tables

Outbox, inbox, fulfillment and stock projection tables

2. RabbitMQ + Worker + Normal Flow

OrderPlacedEvent → outbox → MQ → worker → WMS

3. Resilience under pressure

WMS slow/down, retry, circuit breaker, DLQ, recovery

4. Consistency with POS

Idempotent inbox, sourceVersion, stock projection

5. Traceability + admin view

OrderGuid + messageId + externalRequestId, dashboard

Architectural Evolution of nopCommerce

Scenario C - Omnichannel Commerce Core

TP1 Group 02

Presented by:

- António Alberto 114622
 - Diogo Fernandes 114137
 - João Roldão 113920
 - João Varela 113780
-